



Analysis of the determining factors of good teaching practices of mobile learning at the Spanish University. An explanatory model

Inmaculada Aznar-Díaz, Francisco-Javier Hinojo-Lucena, María-Pilar Cáceres-Reche, José-María Romero-Rodríguez^{*}

University of Granada, Spain

ARTICLE INFO

Keywords:

Mobile learning
Mobile devices
Good teaching practices
Higher education
Evaluation methodologies

ABSTRACT

The use of mobile devices in education is becoming standardized at the higher education stage. However, it is necessary to evaluate good teaching practices of mobile learning (m-learning) in order to set proper application of mobile devices and favourable learning outcomes for students. The purposes of this paper were, on the one hand, to determine the degree of implementation of m-learning and good teaching practices in Spanish universities; on the other hand, to know the causes that lead university professors not to integrate mobile devices and to determine the socio-demographic factors that influence the development of good teaching practices of m-learning. In relation to the researching method, it was used a quantitative approach focused on the implementation of an online survey. A total of 1544 university professors attached to the Faculties of Education throughout Spain, aged between 20 and 77, participated in the study ($M = 45.29$; $SD = 10.45$). The results obtained revealed that: 1) More than 70% of Spanish university professors use mobile devices in the classroom; 2) The main reasons whereby mobile devices are not used in the classroom are ignored, belief that they are a distraction, resistance to change and perceived uselessness; 3) Only 39.56% carried out good teaching practices; 4) The main and possible influential factors good teaching practices of m-learning were professional category, line of research in educational technology, other teaching innovations, believes in the adequacy of mobile devices and the expansion of m-learning in the coming years. Finally, the practical implications of this study are discussed, highlighting the wealth of collected data from this pioneering research on the evaluation of good teaching practices of m-learning.

1. Introduction

Nowadays, it is a well-known fact that, mobile computing has advanced at a relentless pace in recent years. The development of the functionalities of mobile devices has been exponentially, since start making a simple call and sending of Short Message Service (SMS) to the arrival of Internet to the devices and the commerce of mobile applications. This evolution in their functionality has increased the use of mobile devices by educational context as another tool available for teachers. Furthermore, the use of Information and Communication Technologies (ICT) is linked to educational paradigm of the 21st century, where the role of teachers was defined as a learning guide, the main figure to be followed by the students.

^{*} Corresponding author. Faculty of Education Sciences, University of Granada, Campus Universitario Cartuja s/n, 18071, Granada, Spain.

E-mail addresses: iaznar@ugr.es (I. Aznar-Díaz), fhinojo@ugr.es (F.-J. Hinojo-Lucena), caceres@ugr.es (M.-P. Cáceres-Reche), romejo@ugr.es (J.-M. Romero-Rodríguez).

- Sultana, J. (2020). Determining the factors that affect the uses of Mobile Cloud Learning (MCL) platform Blackboard- a modification of the UTAUT model. *Education and Information Technologies*, 25(1), 223–238. <https://doi.org/10.1007/s10639-019-09969-1>.
- Sung, Y. T., Chang, K. E., & Liu, T. C. (2016). The effects of integrating mobile devices with teaching and learning on students' learning performance: A meta-analysis and research synthesis. *Computers & Education*, 94, 252e275. <https://doi.org/10.1016/j.compedu.2015.11.008>.
- Valencia, A., Benjumea, M. L., Morales, D., Silva, A., & Betancur, P. (2018). University professors' attitudes toward the use of mobile devices for academic purposes. *Revista Mexicana de Investigación Educativa*, 23(78), 761–790.
- Vidal, M., Blasco, M. F., & Sastre, M. A. (2019). Determinants of the acceptance of mobile learning as an element of human capital training in organisations. *Technological Forecasting and Social Change*, 149, 119783. <https://doi.org/10.1016/j.techfore.2019.119783>.
- Vygotsky, L. S. (1979). *The development of the higher psychological processes*. Barcelona, Spain: Crítica.
- Xiao-Dong, L., & Hong-Hui, C. (2020). Research on VR-supported flipped classroom based on blended learning — a case study in “learning English through news”. *International Journal of Information and Education Technology*, 10(2), 104–109. <https://doi.org/10.18178/ijiet.2020.10.2.1347>.
- Zheng, L., Li, X., & Chen, F. (2016). Effects of a mobile self-regulated learning approach on students' learning achievements and self-regulated learning skills. *Innovations in Education & Teaching International*, 55(6), 616–624. <https://doi.org/10.1080/14703297.2016.1259080>.
- Zhonggen, Y., & Xiaozhi, Y. (2019). An extended technology acceptance model of a mobile learning technology. *Computer Applications in Engineering Education*, 27, 721–732. <https://doi.org/10.1002/cae.22111>.